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Digital Image Processing

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LINKING EDGE POINTS

Linking Edge Points

- ▶ Ideally, edge detection should only yield sets of pixels on the edges. However, in practice, the edges might not be fully detected because of noise
- ▶ Therefore, after the edge detection, this process is necessary to enhance the edge detection result.

Local Processing

- ▶ One simple approach is to analyse the characteristics of pixels in a small neighbourhood
- ▶ Normally, after edge detections, two properties of gradient will be established: - magnitude and direction
- ▶ Therefore, if a neighbourhood has similar magnitude AND direction (the differences are below thresholds), those two pixels are linked

$$|M(s, t) - M(x, y)| \leq E$$

$$|\alpha(s, t) - \alpha(x, y)| \leq A$$

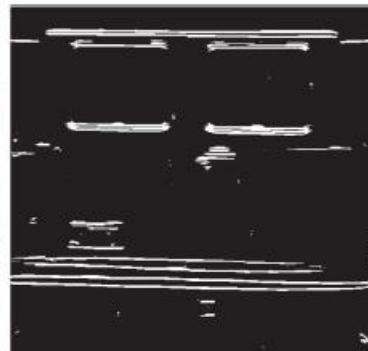
Original image



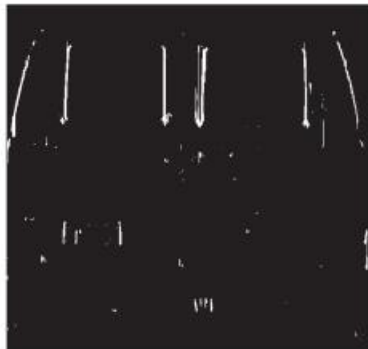
**Gradient
Magnitude**



**Horizontal
Connected Edgels**



**Vertical
Connected Edgels**



**OR set of Horizontal
and Vertical**



**Morphological
Thinning**



Global Processing using Hough Transform

- ▶ The other approach is to accept and eliminate edgels based on predefined global properties of regions of interest.
- ▶ A straight line, with point (x_i, y_i) on, can be generally in a slope-intercept form equation

$$y_i = ax_i + b$$

- ▶ This equation can be altered into ab-plane (parameter space) as follow

$$b = -x_i a + y_i$$

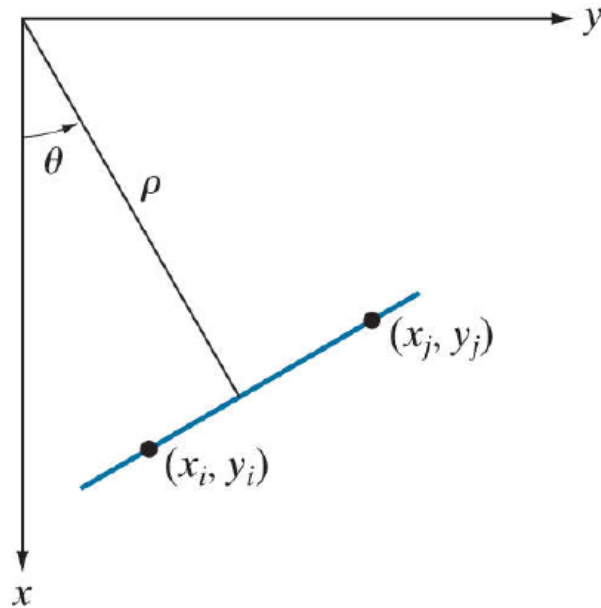
Global Processing using Hough Transform

- ▶ However, in case of vertical lines, the slope is infinite and will be difficult to compute.
- ▶ Using the characteristic of lines where there is another line that normally intersect with. Therefore, we will transform the normal representation of a line in polar form

$$x \cos \theta + y \sin \theta = \rho$$

- ▶ Where ρ is the distance from the origin to the closest point on the straight line and θ is the angle of that closest-point line. This is called the $\rho\theta$ -plane

Global Processing using Hough Transform



Global Processing using Hough Transform

- ▶ Returning to the edge-linking problem, an approach based on the Hough transform is as follows:
 1. Obtain a binary edge map using any of the methods discussed earlier.
 2. Specify subdivisions in the $\rho\theta$ -plane
 3. Examine the counts of the accumulator cells for high pixel concentrations.
 4. Examine the relationship (principally for continuity) between pixels in a chosen cell.

Global Processing using Hough Transform



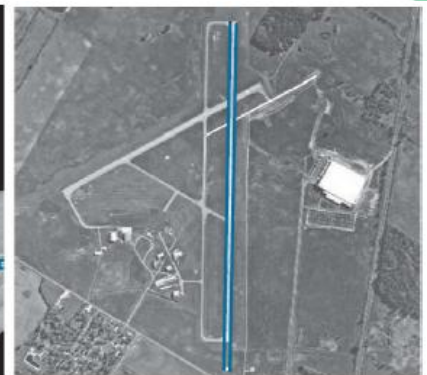
Runway image



Canny edge



**Hough parameter
of 1 degree**



Line of interest

Global Processing using Hough Transform

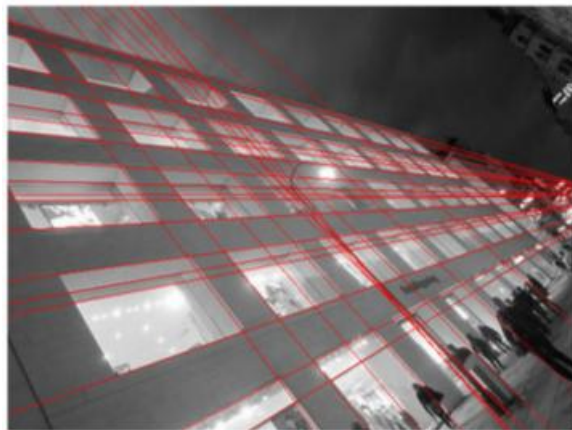
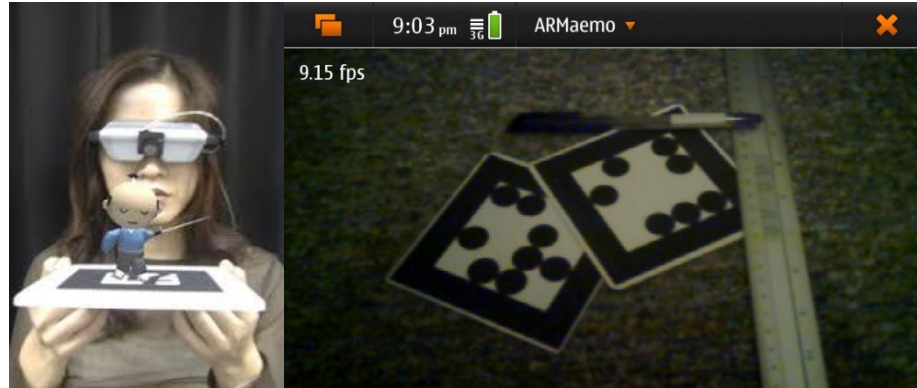


Image from: Manzanera, A., Nguyen, T., & Xu, X. (2016).
Line and circle detection using dense one-to-one Hough transforms on greyscale images.
EURASIP Journal On Image And Video Processing, 2016(1). doi: 10.1186/s13640-016-0149-y

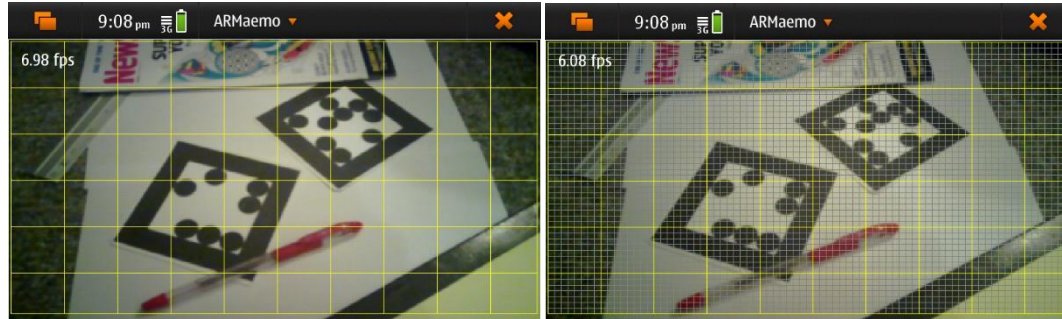
Actual example

- ▶ Markers are needed to be detected. However, in the past, they use flood-fill algorithm to detect the black area which results undetectable markers if they are occluded



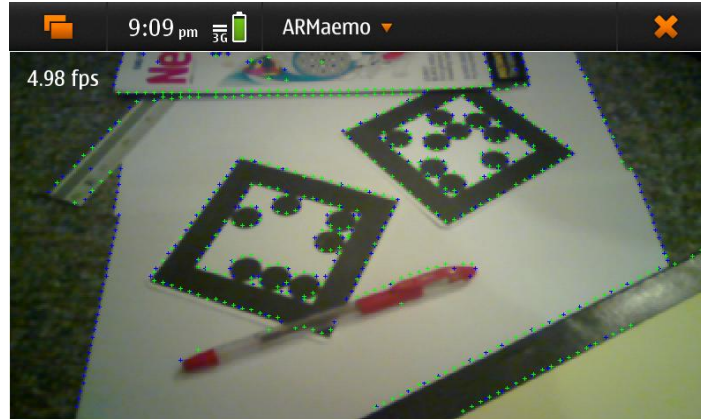
Actual example

- ▶ First, to make it faster to compute, we divide the image into small regions of 40×40 pixels, and divide again into subregions of 5×5 pixels



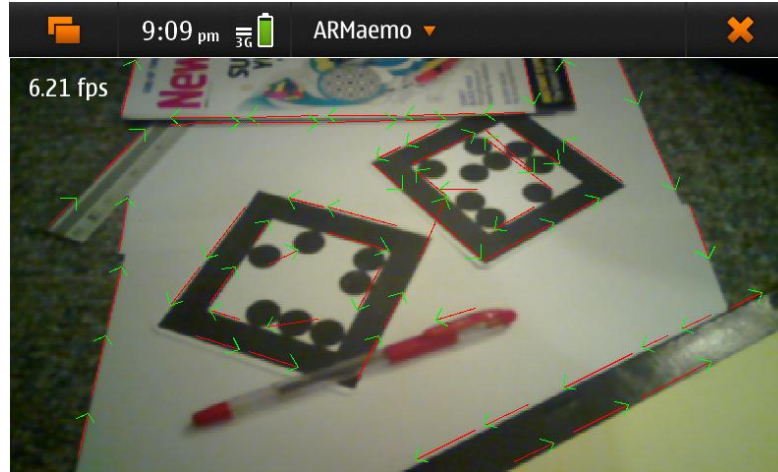
Actual example

- ▶ Smooth the image with Gaussian and using Sobel to detect an edgel from small 5 x 5 subregion. Note that we only need orientations of gradient.. Green dots indicate horizontal edgels and blue dots indicate vertical edgels



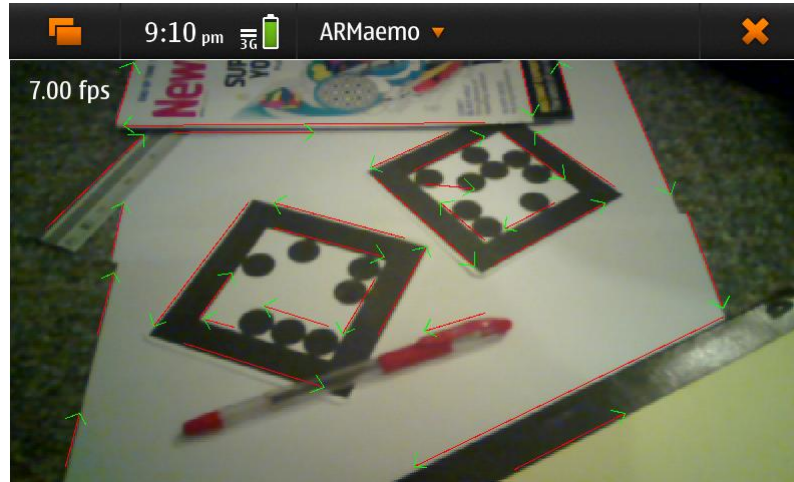
Actual example

- ▶ In the same 40 x 40 region, compare if any two edgels are similar and extend as long as possible within the region. Repeat until we find the longest line with the most support edgels within that region.



Actual example

- ▶ Compare two-line segments if their orientations are similar, close together, and pixels between two lines are on the same edge.



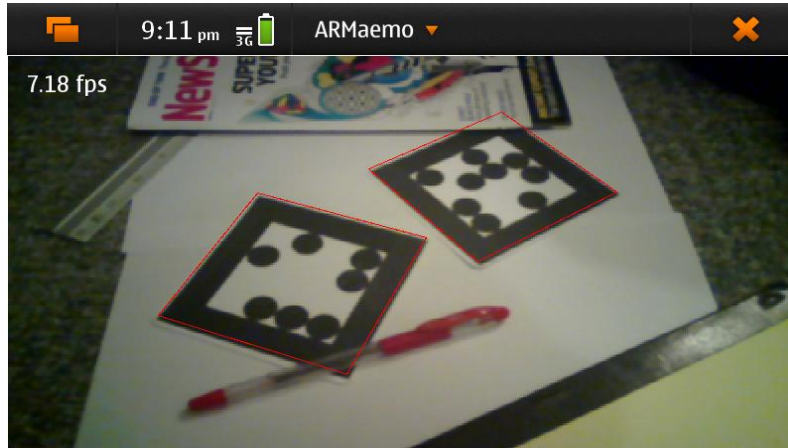
Actual example

- ▶ Extend the line to the corner by continuing detect edgels from the end of each line until it hits corner



Actual example

- ▶ A marker is form of rectangle. Therefore, we look for 4 chains of line that are not in similar orientation and close to each other, using Hough Transform. If exists, extend the lines until they form a rectangle.



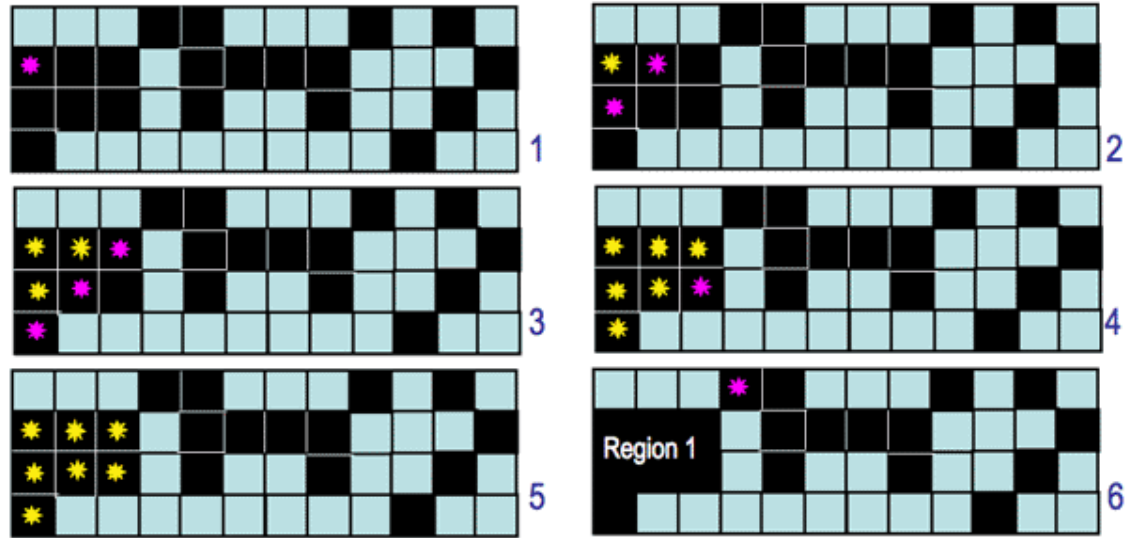


REGION GROWING

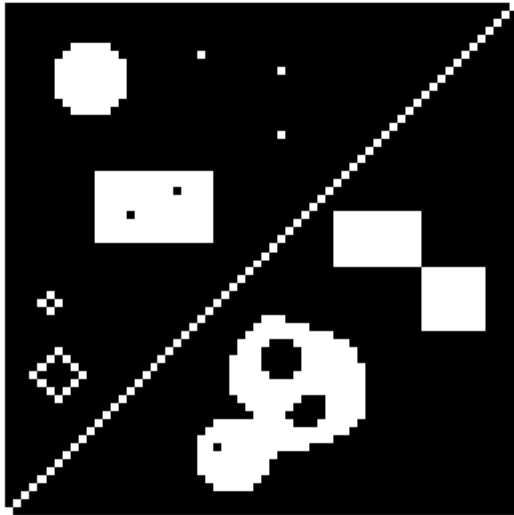
Region Growing

- ▶ Region growing is a bottom-up contextual segmentation which based on predefined criteria.
 - ▶ Connectivity
 - ▶ Similarity
- ▶ There are seed points of the desired regions and the regions will grow according to the criteria above (depends on image).

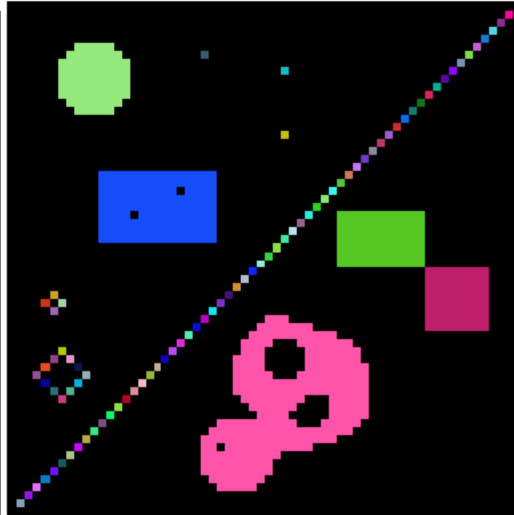
Region Growing



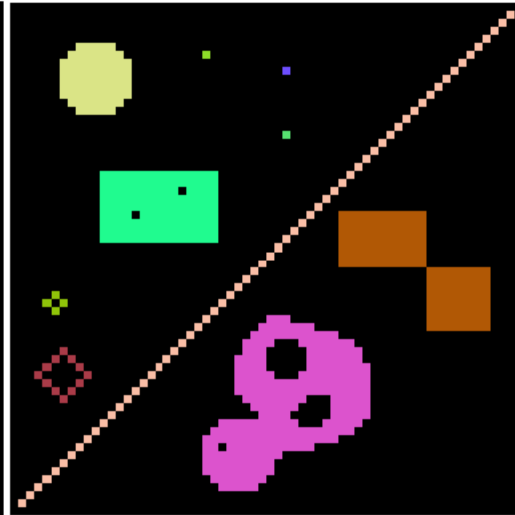
Region Growing



Binary image 64×64 (zoom by a factor of 5)



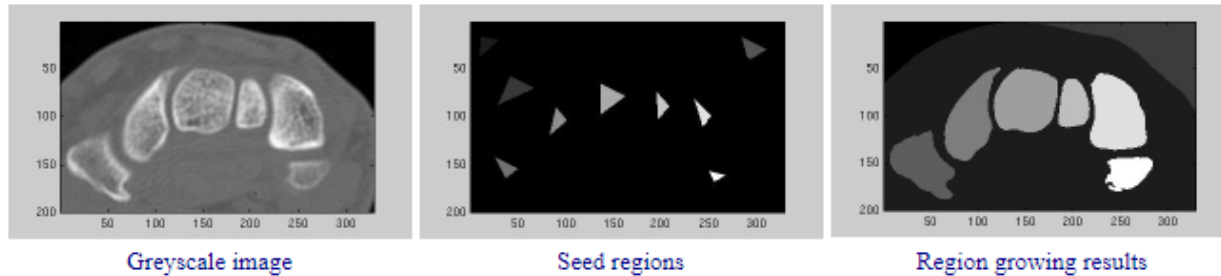
86 foreground 4-connected regions



10 foreground 8-connected regions

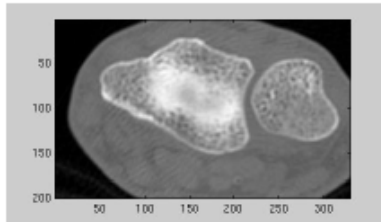
Region Growing

- ▶ The seed points can be set of pixels defined by user which will guarantee that those pixels will be assigned in the same regions

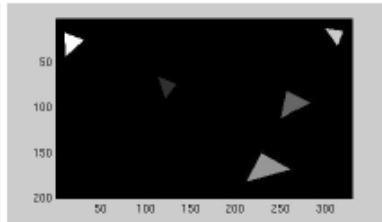


Region Growing

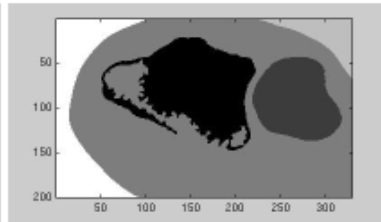
- ▶ This process is simple but unstable as the figure below



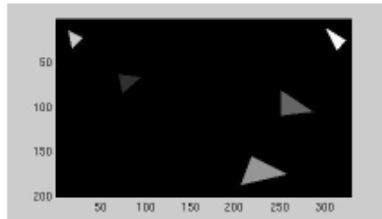
Greyscale image



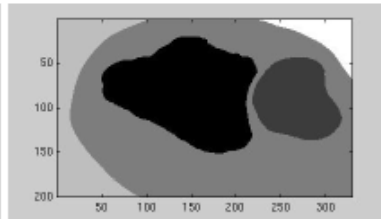
Seed regions: variant 1



Region growing results



Seed regions: variant 2

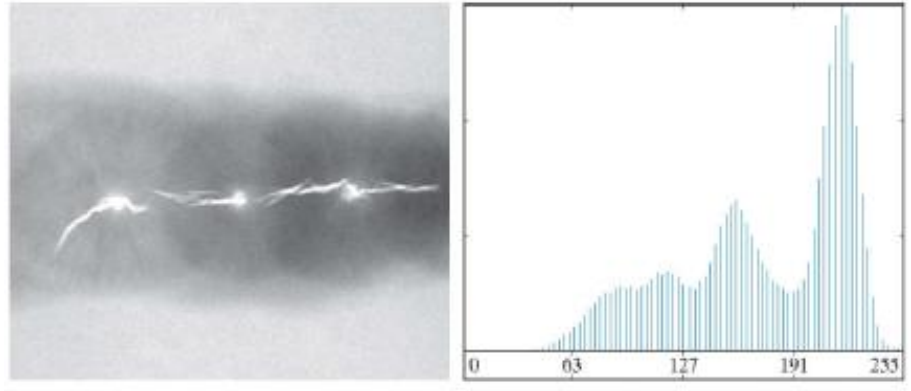


Region growing results

Region Growing

▶ This is an example of region growing of an X-ray image which detect several cracks and porosities

The figures on the right is the X-ray image and its histogram



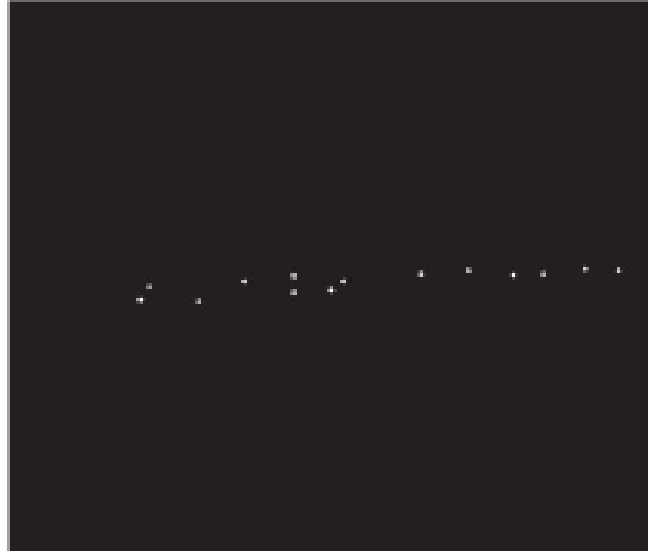
Region Growing

The figure on the right is the thresholded image using $T = 254$, which is the intensity of the crack areas



Region Growing

After repetitive erosion, the regions are reduced to single pixels which will be served as seed points



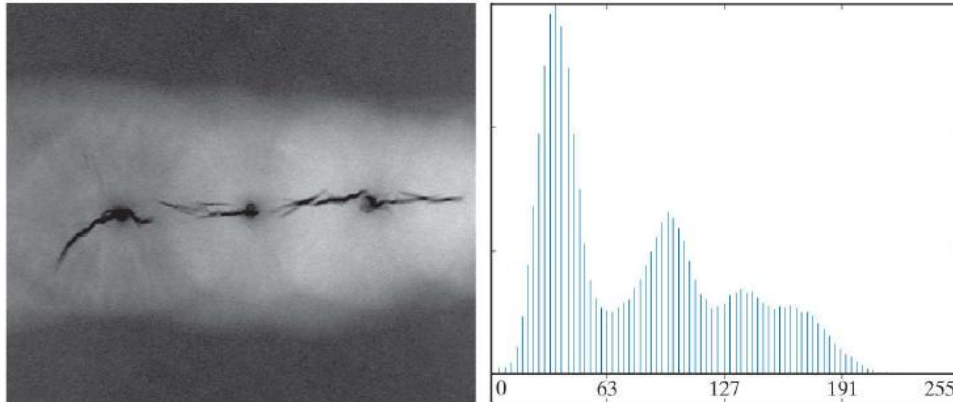
Region Growing

Next, indicate the rule that the similar pixels of each region must have similar intensities. In other word, they must not be different more than a defined threshold value. Therefore, we define the rules as follow.

$$Q = \begin{cases} \text{TRUE} & \text{if the absolute difference of intensities} \\ & \text{between the seed and the pixel at } (x, y) \text{ is } \leq T \\ \text{FALSE} & \text{otherwise} \end{cases}$$

Region Growing

The problem is how to pick T . We can assume that the intensities of the same region should also be in the same PDF. Therefore, using negative image ($254 - \text{intensity of pixel}$), we can see from histogram that T should be 68.



Region Growing

From the seed points, using 8-connected continuity region grow, the region will finally be as follow.





REGION SPLITTING AND MERGING

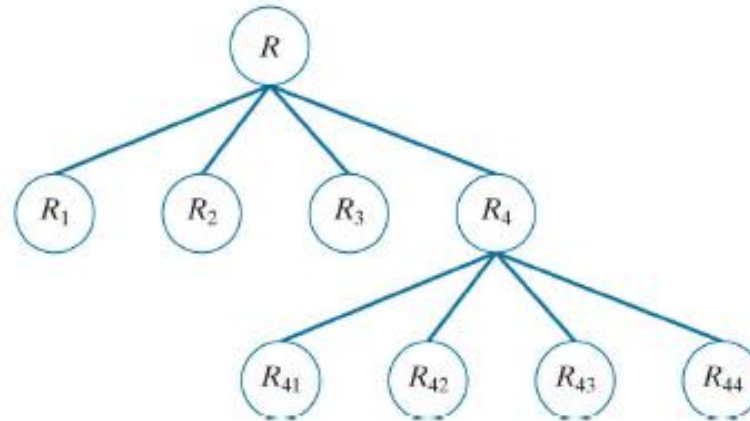
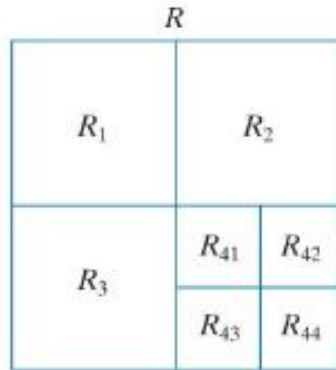
Region Splitting and Merging

- ▶ Region splitting and merging is a top-down contextual segmentation which based on predefined conditions.
- ▶ The idea is, starting from the whole image, if the region does not satisfy the conditions Q , divide that region into quadrants.

$$R_i \rightarrow \begin{cases} \text{divide into 4 regions; if } Q(R_i) = \text{FALSE} \\ \text{remain the same; if } Q(R_i) = \text{TRUE} \end{cases}$$

- ▶ If any of the divided regions still do not satisfy the conditions, the regions will be divided until they satisfy the conditions or hit the limit.
- ▶ Repeat until no splitting possible

Region Splitting and Merging



Region Splitting and Merging

- ▶ After finish splitting, all regions will be checked
- ▶ If $Q(R_j \cup R_k) = TRUE$ and R_j and R_k is two adjacent regions, merge those two regions
- ▶ Repeat until no merging possible

Region Splitting and Merging

This is an image of Cygnus Loop supernova. We want to segment the ring from the rest of image.

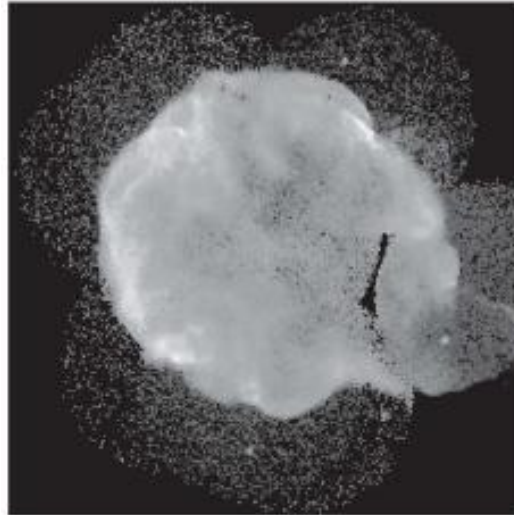
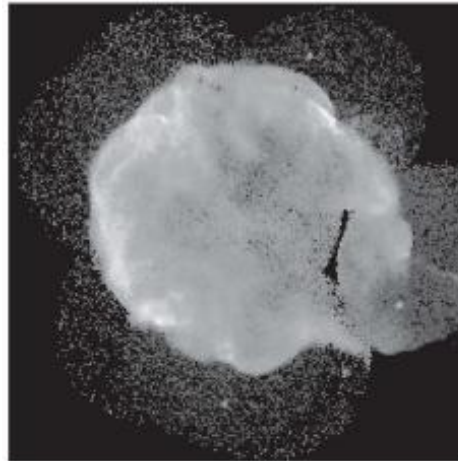


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

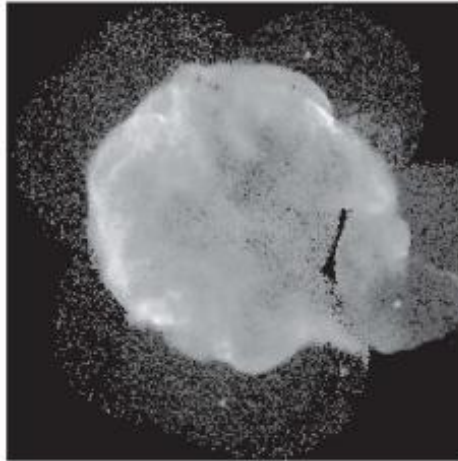
Region Splitting and Merging

The ring should have its average intensity more than the dark area but less than the light inner area



Region Splitting and Merging

The ring has random pixels. Therefore, its standard deviation should be more than outer (dark) and inner (light but smooth) regions



Region Splitting and Merging

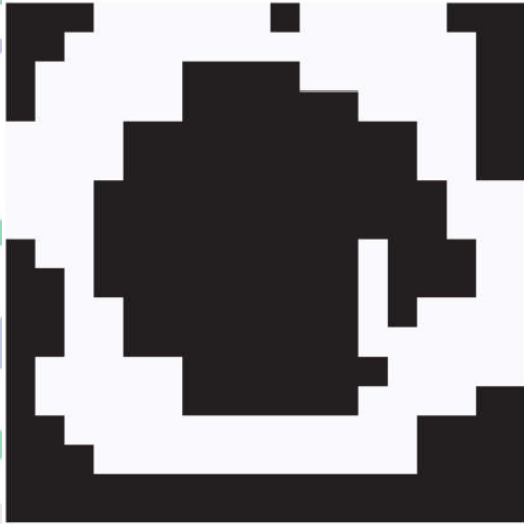
Therefore, the conditions are as follow

$$Q(R) = \begin{cases} \text{TRUE} & \text{if } \sigma_R > a \text{ AND } 0 < m_R < b \\ \text{FALSE} & \text{otherwise} \end{cases}$$

where σ_R and m_R are the standard deviation and mean of the region being processed, and a and b are nonnegative constants.

By analysing the ring region, its mean intensity does not exceed 125 and standard deviation is always greater than 10

Region Splitting and Merging



**Split and merge with smallest
allowed quadregion = 32x32**



**Split and merge with smallest
allowed quadregion = 16x16**



**Split and merge with smallest
allowed quadregion = 8x8**



End of Week 10